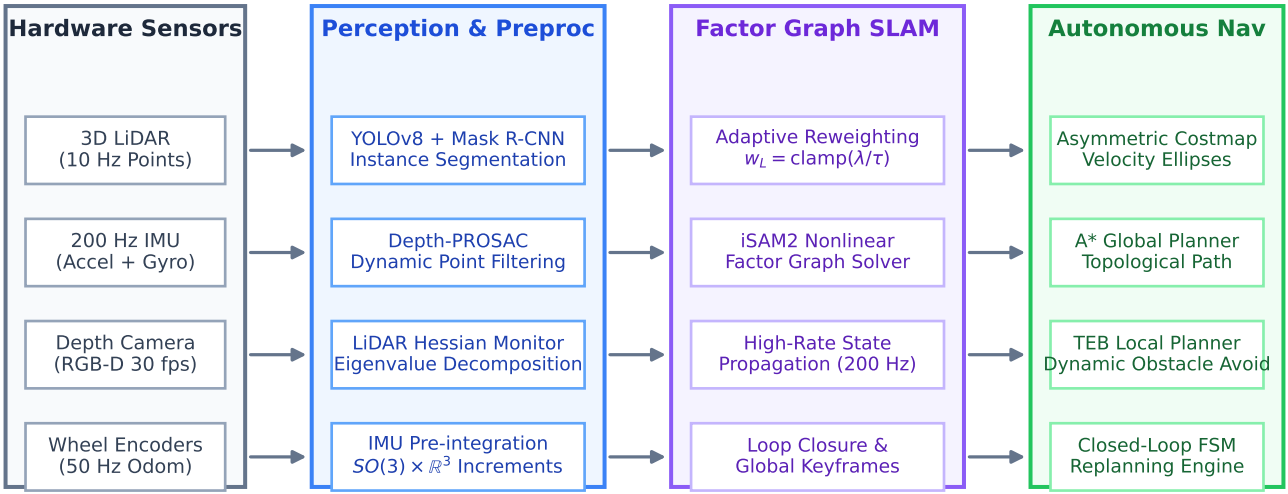
Hardware Sensors
3D LiDAR
(10 Hz Points)
200 Hz IMU
(Accel + Gyro)
Depth Camera
(RGB-D 30 fps)
Wheel Encoders
(50 Hz Odom)
Perception & Preproc
YOLOv8 + Mask R-CNN
Instance Segmentation
Depth-PROSAC
Dynamic Point Filtering
LiDAR Hessian Monitor
Eigenvalue Decomposition
IMU Pre-integration
$SO(3) \times \mathbb{R}^3$ Increments
Factor Graph SLAM
Adaptive Reweighting
$w_L = \text{clamp}(\lambda/\tau)$
iSAM2 Nonlinear
Factor Graph Solver
High-Rate State
Propagation (200 Hz)
Loop Closure &
Global Keyframes
Autonomous Nav
Asymmetric Costmap
Velocity Ellipses
A* Global Planner
Topological Path
TEB Local Planner
Dynamic Obstacle Avoid
Closed-Loop FSM
Replanning Engine